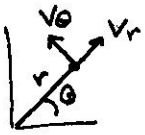


Consider the fluid flow that derives from the potential

$$\varphi = \frac{Q}{2\pi} \ln r + U_{\infty} r \cos \theta,$$

corresponding to the superposition of a uniform stream of velocity U_{∞} and a source of volumetric flux Q .

1. Determine the velocity components in cartesian and cylindrical coordinates.
2. Obtain the stream lines.
3. Compute the trajectories and paths.
4. Calculate the stagnation points, sketching the stream lines that reach them.
5. Consider the flow near the stagnation points, determining in particular the trajectories and the stream lines.

$$1) \quad \begin{aligned} V_r &= \frac{\partial \varphi}{\partial r} = \frac{Q}{2\pi} \frac{1}{r} + U_{\infty} \cos \theta \\ V_{\theta} &= \frac{1}{r} \frac{\partial \varphi}{\partial \theta} = -U_{\infty} \sin \theta \end{aligned} \quad \left| \quad \begin{aligned} V_x &= V_r \cos \theta - V_{\theta} \sin \theta = \frac{Q}{2\pi} \frac{x}{(x^2+y^2)} + U_{\infty} \\ V_y &= V_r \sin \theta + V_{\theta} \cos \theta = \frac{Q}{2\pi} \frac{y}{x^2+y^2} \end{aligned} \right.$$


FOR ANY CLOSED LINE SURROUNDING THE ORIGIN $\oint \vec{v} \cdot \vec{n} d\vec{s} = Q$ e.g. $\oint \vec{v} \cdot \vec{n} d\vec{s} = \int_0^{2\pi} V_r r d\theta = Q$

$$2) \quad \frac{dr}{\frac{Q}{2\pi} \frac{1}{r} + U_{\infty} \cos \theta} = \frac{r d\theta}{-U_{\infty} \sin \theta} \Rightarrow \frac{Q}{2\pi} d\theta + U_{\infty} r \cos \theta d\theta + U_{\infty} \sin \theta dr = 0$$

$$d\left(\frac{Q\theta}{2\pi} + U_{\infty} \sin \theta r\right) = 0 \Rightarrow \boxed{\frac{Q\theta}{2\pi} + U_{\infty} \sin \theta r = \frac{Q\theta_0}{2\pi} + U_{\infty} \sin \theta_0 r_0}$$

$$3) \quad \frac{dr}{dt} = \frac{Q}{2\pi} \frac{1}{r} + U_{\infty} \cos \theta \quad \left| \quad \begin{aligned} t &= 0 \\ r &= r_0 \\ \theta &= \theta_0 \end{aligned} \right. \quad \text{DIVIDING AND INTEGRATING WE OBTAIN THE PATH LINES}$$

$$r \frac{d\theta}{dt} = -U_{\infty} \sin \theta$$

$$dt = -\frac{r d\theta}{U_{\infty} \sin \theta} = -\frac{\frac{Q}{2\pi} (\theta_0 - \theta) + U_{\infty} \sin \theta_0 r_0}{U_{\infty}^2 \sin^2 \theta} d\theta$$

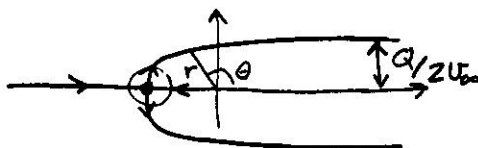
$$t = -\int_{\theta_0}^{\theta} \frac{\frac{Q}{2\pi} (\theta_0 - \theta) + U_{\infty} \sin \theta_0 r_0}{U_{\infty}^2 \sin^2 \theta} d\theta$$

$$4) \quad V_{\theta} = 0 \rightarrow \theta = 0 \text{ or } \theta = \pi \quad V_r = 0 \rightarrow r = -\frac{Q}{2\pi U_{\infty} \cos \theta}$$

THERE EXISTS ONE STAGNATION POINT GIVEN BY

$$\boxed{\theta = \pi, \quad r = \frac{Q}{2\pi U_{\infty}} = r_0}$$

TWO STREAMLINES REACH THE STAGNATION POINT: $\theta = \pi, 0 < r < \infty$



5) IN TERMS OF THE LOCAL VARIABLES DIMENSIONLESS

$$\xi = \frac{x - r_0}{r_0} < 1$$

$$\eta = \frac{y}{r_0} < 1$$

$$\Rightarrow V_{\xi} = \frac{V_x}{U_{\infty}} = -\xi$$

$$V_{\eta} = \frac{V_y}{U_{\infty}} = \eta$$

$$z = \frac{t}{r_0/U_{\infty}}$$

$$r = \frac{Q}{2\pi} \frac{(\pi - \theta)}{U_{\infty} \sin \theta}, \quad 0 < \theta < 2\pi$$

$$\hookrightarrow (\theta \rightarrow 0, r \sin \theta = \frac{Q}{2U_{\infty}})$$

STREAMLINES $\frac{d\xi}{-\xi} = \frac{d\eta}{\eta} \Rightarrow \boxed{\xi \eta = \xi_0 \eta_0}$

TRAJECTORIES

$$\frac{d\xi}{dz} = -\xi$$

$$\frac{d\eta}{dz} = \eta$$

$$\boxed{\xi = \xi_0 e^{-z}, \quad \eta = \eta_0 e^z}$$